

CORE CURRICULUM



Financial Accounting

V.G. Narayanan and Dennis Campbell, Co-Series Editors

READING

+ INTERACTIVE ILLUSTRATIONS

Analyzing Financial Statements

.....
V.G. NARAYANAN

HARVARD BUSINESS SCHOOL

SURAJ SRINIVASAN

HARVARD BUSINESS SCHOOL
.....

5056 | Published: April 11, 2017

Table of Contents

1 Introduction	3
2 The Financial Analysis Framework	4
2.1 Understanding Return on Equity	6
2.2 Profitability	8
2.3 Operating Efficiency	11
2.3.1 Return on Assets	12
2.3.2 Digging Deeper into Asset Turnover	13
2.3.3 Inventory Turnover	14
2.3.4 Accounts Receivable Turnover	16
2.3.5 Accounts Payable Turnover	18
2.3.6 Cash Conversion Cycle	20
2.3.7 Working Capital Turnover	22
2.4 Financial Leverage	23
2.4.1 Interest Coverage Ratio	27
2.4.2 Liquidity Ratios	28
3 Summary	30
4 Supplemental Reading	32
4.1 Key Ratios	32
5 Key Terms	34
6 Endnotes	36
7 Index	37



This reading contains links to online interactive illustrations, denoted by the icon above. To access these exercises, you will need a broadband Internet connection. Verify that your browser meets the minimum technical requirements by visiting <http://hbsp.harvard.edu/tech-specs>.

V.G. Narayanan, Thomas D. Casserly, Jr. Professor of Business Administration, Harvard Business School, and Suraj Srinivasan, Philip J. Stomberg Professor of Business Administration, Harvard Business School, developed this Core Reading with the assistance of writer M. Penelope K. Rossano.

Copyright © 2017 Harvard Business School Publishing Corporation. All rights reserved.

1 INTRODUCTION

The ability to analyze financial statements is essential for anyone trying to understand a business and assess its performance. Managers can use insights derived from financial analysis to inform their strategic decisions, and outside stakeholders, such as debt and equity investors, can use the information to evaluate a company's performance. For example, managers might want to assess an acquisition's performance implications, a bank or supplier would need to assess a client's creditworthiness before granting a loan or extending credit, and customers would like to understand a potential supplier's long-term viability before entering into a contract. In this reading, we will introduce financial analysis tools that can be used to gain insights about a company's business model and financial performance. We will primarily use a ratio analysis method called the DuPont framework. In this framework, one ratio tells us how efficiently a company is utilizing its assets, while another ratio compares the company's cash to its upcoming liabilities to see if it will likely be able to pay its creditors. Taken together, these ratios can be used to understand and assess the company's financial and operational performance.

Ratio analysis is useful in making comparisons across companies or in evaluating a company's performance over time. In the former case, financial analysis can be used to assess the performance of managers across similar companies. In the latter case, time-series analysis allows us to assess how well managers are executing the desired strategy over time.

Throughout this reading, we will use the publicly available financial statements of several retail companies to provide examples of various types of ratio analyses and comparisons. They are Industria de Diseño Textil, S.A (Inditex Group), a Spanish fast-fashion retailer; Prada S.p.A. (Prada), a luxury retail brand with Italian roots and headquarters in Hong Kong; and three US-headquartered brands: The Gap, Inc.; Nordstrom, Inc.; and Urban Outfitters, Inc.

In 2015, Inditex was operating worldwide with over 6,600 stores¹ of fast-fashion brand chains such as Zara, Pull & Bear, Stradivarius, and Massimo Dutti. It was known in the retail world for being innovative in managing its supply chain and for going very quickly from design to garment in about six weeks.

Founded in 1913 in Milan, by 2015 Prada was a luxury brand that produced and sold high-quality leather goods, clothing, footwear, and accessories including eyewear and fragrance for both men and women. It was operating internationally in more than 70 countries at more than 600 directly operated stores at prestigious locations.² Prada executives carefully monitored both the creative and production processes to guarantee excellent quality and exclusivity in its products. The company's business strategy combined its in-house design skill and industrial know-how to create inventive, high-quality, trend-setting products.³

The Gap, Inc., was a global retailer offering apparel, accessories, and personal products for men, women, and children under Gap, Banana Republic, Old Navy, Athleta, and Intermix brands. In 2015, the company operated more than 3,700 company-owned and franchise locations. Its brand appeal was targeted at younger customers.

In 2015, Urban Outfitters, Inc., operated under the Urban Outfitters, Anthropologie, Free People, Terrain, and BHLDN brands.⁴ The company targeted a broad range of culturally sophisticated customers: young adults aged 18–28 at Urban Outfitter stores, women aged 28–45 at Anthropologie, and young, contemporary women aged 25–30 at private label Free People.⁵

Nordstrom, Inc., founded in 1901 as a shoe store in Seattle, was committed to providing superior customer service and delivering the best possible shopping experience. In 2015, it was a leading fashion specialty retailer with an established e-commerce business. The company strived to maintain its reputation for its high level of integrity, excellent customer service, and quality merchandise by offering an extensive selection of high-quality, brand-name, private-label merchandise.⁶

2 THE FINANCIAL ANALYSIS FRAMEWORK

Financial statement information comes from the three key statements: the income statement, the balance sheet, and the statement of cash flows. The **income statement** documents the company's financial activity over a given accounting period, say, a year. The **balance sheet**, on the other hand, reflects the company's financial position, assets, liabilities, and shareholder equity as of a particular date, say, the end of the financial year. Like the income statement, the **statement of cash flows** also presents

the firm's cash flows over a given period, such as a year. In calculating ratios, it is common to average the beginning and ending balance sheet amounts to measure the average level of assets used during the period. We use this convention because income statement items (e.g., revenues) are measured over the course of the year, so we assume that the average balance sheet values have been used throughout the year to generate a given level of revenues. However, some financial statement users may use beginning-of-the-year amounts if appropriate, such as when the ratios will be used for forecasting. Throughout this reading, we will use the average balance sheet amounts when comparing income statement and balance sheets in any ratio. Certain ratios may also be computed for shorter periods, such as quarters or months. For example, to plan inventory levels, it is important to understand seasonal cycles, so inventory ratios are often computed on a quarterly or monthly basis.

When conducting financial analysis, key questions an analyst typically asks relate to the company's performance. For example, how well has the company executed its strategy? How well has it performed against its competitors? Does the performance arise from superior operating capability or because of its financing mix? We will begin our financial analysis of the company using two key metrics of performance: **return on equity (ROE)** and **return on assets (ROA)**. Both of these metrics allow an analyst to assess how well a company has generated profits from the resources deployed.

ROE measures overall business performance, specifically management's ability to generate profits for its shareholders. It is calculated by dividing net income by total **shareholders' equity (SE)**.

ROA, also referred to as "return on investment," is an indicator of the company's profitability relative to its assets or total capital employed in the firm (recall that assets = liabilities + equity); it indicates how efficiently management uses the company's assets to generate earnings. ROA is calculated by dividing a company's earnings by its total assets.

As we will see later, breaking down each of these ratios into its underlying components allows us to assess different aspects of a company's performance relative to comparable companies as well as to the firm's past results.

The **DuPont framework**, commonly used for financial analysis, originated at the DuPont Corporation in the 1920s. It decomposes ROE into subcomponents to provide a deeper look at how the profit (or loss) was generated. Systematic analysis of the ratios that make up the DuPont framework provide additional information about how a company manages its business. Similar to peeling an onion, conducting financial statement analysis that examines the most general ratios, then drills down to its more granular components facilitates the analyst's understanding of the determinants of performance. For example, an analyst may ask whether a company's

performance is primarily driven by profit margins (how much profit the company generates for its shareholders), asset turnover (how efficiently the company uses its assets to generate return), or **financial leverage** (the level of debt the company carries).

2.1 Understanding Return on Equity

Let's start with the return on equity, which can be broken into three components:

- Profitability
- Operating efficiency
- Financial leverage

These values are derived by decomposing ROE into three parts:

$$\text{ROE} = \text{net profit margin} \times \text{asset turnover} \times \text{financial leverage}$$

That is,

$$\text{ROE} = \frac{\text{net income}}{\text{sales}} \times \frac{\text{sales}}{\text{total assets}} \times \frac{\text{total assets}}{\text{equity}}$$

Which simplifies to,

$$\begin{aligned}\text{ROE} &= \frac{\text{net income}}{\text{sales}} \times \frac{\text{sales}}{\text{total assets}} \times \frac{\text{total assets}}{\text{equity}} \\ &= \frac{\text{net income}}{\text{equity}}\end{aligned}$$

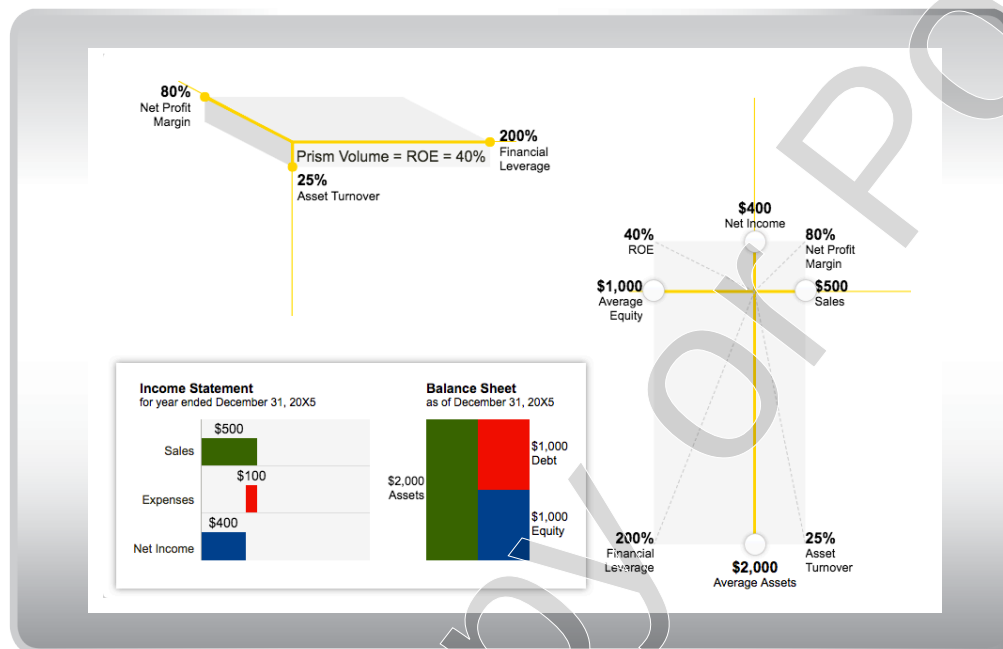
Interactive Illustration 1 demonstrates in a graphical format how the three components of ROE link together. Change the numerical values of net income, sales, assets, and equity, and observe their effects on the ROE “block,” as well as on the simplified income statement and balance sheet.



INTERACTIVE ILLUSTRATION 1 DuPont Framework



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2pHtruh



Managers can use the financial analysis framework to evaluate the sources of favorable or unfavorable performance in each of these three areas, which helps them take actions to increase the company's overall ROE. In this reading, to demonstrate how each of the underlying factors contributes to a company's ROE, we will calculate each ratio for Prada and then provide the ratios for the other companies described in the introduction. Prada's financial statement numbers are given in **Exhibit 1**. All financial numbers are in millions of euros.

Note that we average the beginning and ending balances of shareholders' equity for the period in which net income is earned for our analysis. To illustrate, we will plug in the values from Exhibit 1, as follows:

$$\begin{aligned}
 &\text{Prada's average shareholders' equity (SE) in 2015} \\
 &= \frac{(\text{beginning SE} + \text{ending SE})}{2} \\
 &= \frac{(\text{€2,688} + \text{€3,001})}{2} \\
 &= \text{€2,845}
 \end{aligned}$$

$$\text{ROE for Prada in 2015} = \frac{\text{€451}}{\text{€2,845}} = 15.8\%$$

Similarly, we calculate ROE for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
ROE	41.8%	25.4%	31.9%	15.8%	15.4%

Note that the ROEs for Prada and Urban Outfitters differ by less than half a percentage point. Gap is highly profitable, with ROE of 41.8%. Further analysis of the drivers of ROE will inform us about how these companies manage their business.

2.2 Profitability

The first component of the ROE decomposition is **profitability**, the profit margin that the company achieves from each dollar of sales after all expenses have been accounted for:

$$\text{Net profit margin} = \frac{\text{net income}}{\text{sales}}$$

Using the values from Exhibit 1, we find:

$$\text{Net profit margin for Prada in 2015} = \frac{\text{€451}}{\text{€3,552}} = 12.7\%$$

Similarly, we calculate the net profit margin for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Net Profit Margin	7.7%	13.8%	5.3%	12.7%	7.0%

The net profit margin for Urban Outfitters, 7%, and Prada, almost 13%, indicates that Prada would earn about twice Urban Outfitters' profit for every dollar of sales after accounting for all costs. This margin structure fits their business models. Prada is a luxury brand that sells high-end specialty products, while Urban Outfitters' products appeal to more cost-conscious customers. Gap, whose net profit margin of 7.7% is similar to that of Urban Outfitters, also has a similar customer profile. Urban Outfitters and Gap are both low-cost providers, whereas Prada's more differentiated model operates at the higher end of the customer spectrum. Companies with a low-

cost model tend to sell larger quantities at lower margins, while companies with a differentiated strategy tend to have higher profit margins but sell lower volumes. Recall that the ROE for both Urban Outfitters and Prada is approximately 15%. We will discuss later what may account for this similarity in ROE despite the big difference in the two companies' net profit margin.

A big advantage of the DuPont decomposition framework is that it lets us dig deeper into the components of ROE ("peeling the onion" analogy again), in this case, net income margin. For example, some investors may be interested in the **gross profit margin**, which represents the amount of profit that is left to cover other expenses after only the cost of goods sold is subtracted from revenues. This means that gross profit margin measures profit before operating expenses as a percent of sales. It is calculated by dividing the net revenue minus cost of goods sold, by sales.

$$\text{Gross profit margin} = \frac{\text{gross profit}}{\text{sales}}$$

Using the data from Exhibit 1, we get the following:

$$\text{Gross profit margin for Prada in 2015} = \frac{\text{€2,551}}{\text{€3,552}} = 71.8\%$$

Similarly, we calculate gross profit margin for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Gross Profit Margin	38.3%	58.3%	36.9%	71.8%	35.4%

As discussed earlier, Prada is a luxury brand; a gross margin at 72% is consistent with the higher price of its products. Urban Outfitters, on the other hand, is targeting a customer group of younger adults, so the lower gross margin of 35% will facilitate a more affordable price to their customers.

Examining a company's expense structure can also help you better understand its performance. One common performance metric for a retail company is how efficient its selling, general, and administrative expenses (SG&A) are. **SG&A margin** is defined as the amount of SG&A expenses incurred by a company for every dollar of revenues earned. SG&A expenses are one component of operating expenses.

$$\text{SG\&A margin} = \frac{\text{selling, general, and administrative (SG\&A) expenses}}{\text{sales}}$$

which, using the data in Exhibit 1, gives us the following:

$$\text{Prada's SG\&A margin in 2015} = \frac{\text{€1,716}}{\text{€3,552}} = 48.3\%$$

Similarly, we calculate SG&A margin for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
SG&A Margin	25.7%	35.6%	27.2%	48.3%	24.4%

Since SG&A includes expense items such as advertising and selling expenses, it is appropriate for Prada to have a higher expense level than Urban Outfitters and Gap given Prada's higher-end advertising and likely greater selling expenses for its luxury brand compared to the fast-fashion retailers.

Review how these common profitability measurements are derived from the income statement in **Interactive Illustration 2**. Select one of the four metrics at the top of the graphic, and explore how those metrics would be affected by different performance scenarios (e.g., more annual revenue, reduced costs, etc).



INTERACTIVE ILLUSTRATION 2 Profitability Ratios^a



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2pHtruh



^a For simplicity, this representation limits the range of possible values. For example, COGS, and operating expenses could both be larger than revenue, leading to a net loss.

2.3 Operating Efficiency

In addition to questions about profitability, analysts also often ask how efficiently a company is using its assets. The next component of the ROE decomposition, asset turnover, helps answer this question. **Asset turnover** tells us the extent of sales generated by a dollar (or euro in our example) of assets. The amount of sales generated by a unit of assets depends on the strategy the company uses to generate sales or deliver a service. This efficiency measure also depends on the technology the company is using. For instance, a retailer with greater percentage of online sales—ones where fixed assets are not needed to deliver revenues—will have a greater asset turnover than a purely brick-and-mortar retailer.

$$\text{Asset turnover} = \frac{\text{sales}}{\text{average total assets}}$$

Applying this to our Prada example gives us the following:

$$\begin{aligned} &\text{Prada's average total assets in 2015} \\ &= \frac{(\text{beginning total assets} + \text{ending total assets})}{2} \\ &= \frac{(\text{€3,888} + \text{€4,739})}{2} \\ &= \text{€4,314} \end{aligned}$$

$$\text{Prada's asset turnover in 2015} = \frac{\text{€3,552}}{\text{€4,314}} = 0.8$$

Similarly, we calculate asset turnover for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Asset Turnover	2.1	1.2	1.5	0.8	1.6

It is worthwhile to repeat why we have used average total assets rather than the beginning or ending value of assets in the denominator. A company records its sales at every point throughout the year using assets in place. Assets in place may change during the year depending on the company's investments. Averaging the start- and end-of-year assets provides a basis for comparison when looking at assets that support the level of sales achieved over the year.

Urban Outfitters' asset turnover is twice that of Prada's, meaning the fast-fashion retailer sells twice as much per unit of assets as the more luxury-oriented brand. This is likely because Prada has to make larger investments for a given level of sales. Prada stores need to be located in more upscale areas and have more fashionable interiors than Urban Outfitters does. The value of inventory in a Prada store is also likely to be greater than that in an Urban Outfitters store. The difference in business models is apparent in the ratios: Prada, with its higher-end luxury brands, generates greater profit per unit of revenue than Urban Outfitters, but sells lower volume for the assets in place. Recall that the net profit margin for Prada was about twice that of Urban Outfitters, meaning that Urban Outfitters makes lower profit per unit of revenue but sells twice as much as Prada does per unit of assets that it uses. These two factors net each other out when we compare the two companies' profitability per unit of asset utilized. This brings us to our next concept, return on assets.

2.3.1 Return on Assets

Return on assets (ROA) measures profits as a proportion of total resources used or financed by the firm. It is calculated by dividing net income by average total assets. ROA is the product of the first two terms of the DuPont decomposition: net profit margin $\times$ asset turnover.

$$\text{ROA} = \frac{\text{net income}}{\text{average total assets}}$$

Using our example, we find the following:

$$\text{Prada's ROA in 2015} = \frac{\text{€451}}{\text{€4,314}} = 10.4\%$$

Similarly, we calculate ROA for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
ROA	16.2%	17.2%	8.1%	10.4%	11.3%

As discussed earlier, while Prada and Urban Outfitters have distinctly different net profit margins and asset turnover ratios consistent with their differing market positions and business strategies, when we consider ROA, the two effects almost cancel each other out. Note here that Nordstrom's ROA is 8.1%, which is somewhat lower than Prada's 10.4%. Despite the lower ROA, Nordstrom's ROE in the ROE

table in Section 2.1 is much higher, 31.9%, compared to Prada's 15.8%. We will return to this apparent puzzle later when we discuss leverage.

A firm's total assets are financed by a mixture of debt and equity. Net income (the numerator in the ROA formula) includes interest expense, which is the financing cost of debt. If the analyst wishes to ignore the firm's financing mix, he or she would eliminate the financing impact on net income, which would determine a pure return on investment before financing costs. ROA is often calculated by dividing after-tax net income before interest expense by average total assets.

2.3.2 Digging Deeper into Asset Turnover

Companies need a mix of assets to drive their business performance. For example, retail companies need stores (long-term assets) as well as shorter-term assets such as inventories. By examining whether asset turnover efficiency is driven by short-term or long-term assets, an analyst can learn about how well managers are undertaking long-term investments versus how well they are handling the business's day-to-day operational activities.

$$\text{Long-term asset turnover} = \frac{\text{sales}}{\text{average long-term assets}}$$

Here are the long-term asset turnover numbers for our retail examples:

$$\begin{aligned} &\text{Prada's average long-term (LT) assets in 2015} \\ &= \frac{(\text{beginning LT assets} + \text{ending LT assets})}{2} \\ &= \frac{(\text{€2,427} + \text{€2,839})}{2} \\ &= \text{€2,633} \end{aligned}$$

$$\text{Prada's LT asset turnover in 2015} = \frac{\text{€3,552}}{\text{€2,633}} = 1.3$$

Similarly, we calculate LT asset turnover for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
LT Asset Turnover	4.8	2.4	3.7	1.3	2.9

As with the total asset turnover ratios, Urban Outfitters uses its LT assets more efficiently than Prada does. Gap appears to be the most efficient in utilizing its long-term assets to produce sales revenues.

Let's turn to short-term operations. A company needs **working capital** to manage its day-to-day operations. Working capital is composed of receivables plus inventory less the amount the company owes to its vendors, such as accounts payable. Companies need smaller amounts of working capital if they can collect more quickly from their customers (lower receivables), hold lower levels of inventory for a given level of sales (faster inventory turnover), and receive greater financing from their vendors (greater accounts payable). Let's examine each of these in turn.

2.3.3 Inventory Turnover

Inventory turnover, calculated as cost of goods sold divided by the average inventory for the period, measures how often the inventory is sold during a given time period. It is useful in understanding how efficiently a business manages its inventory levels. Just as holding more assets for a given level of sales does, holding more inventory for a given level of sales decreases efficiency. Although businesses may prefer to hold lower inventory levels, the trade-off is to avoid running out of products available for customers. Inventory levels also depend on product characteristics (e.g., perishable products versus packaged goods), stocking efficiency, variation in product demand, and so forth. As when we calculate asset turnover, we use the average inventory balance rather than the ending balance. This is especially relevant for a firm that is growing quickly, as the level of inventory may have increased significantly during the year.

$$\text{Inventory turnover} = \frac{\text{cost of goods sold (COGS)}}{\text{average inventory}}$$

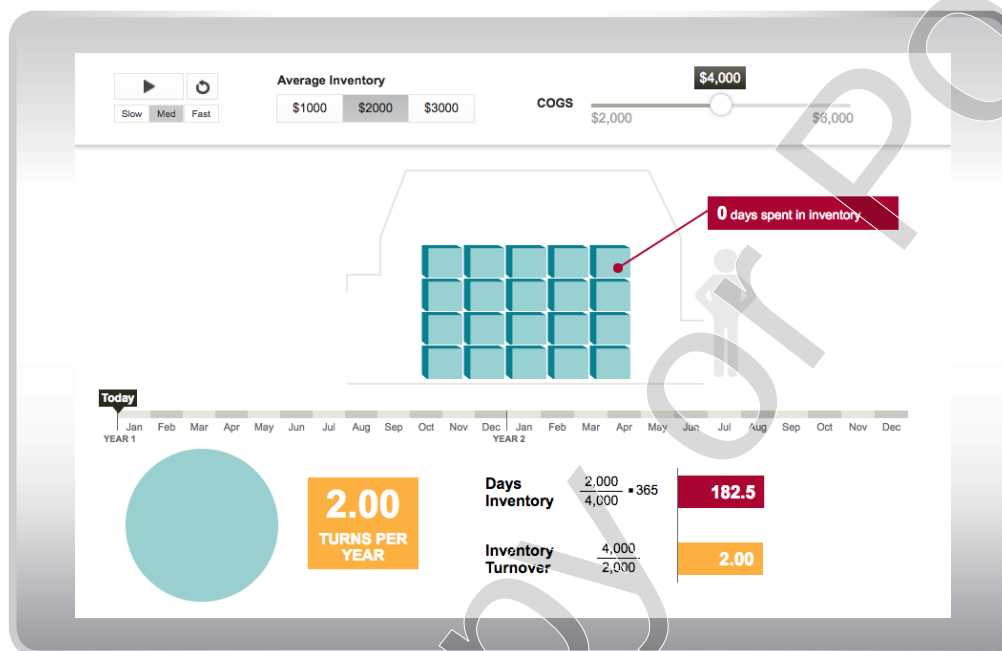
Interactive Illustration 3 is a demonstration of inventory turnover in action. Start the animation of items moving through a warehouse facility over the course of two years of business operations. Reset the illustration and replay it with different assumptions about the firm's average inventory and annual COGS. The effect of these parameters are reflected in the "Days Inventory" and "Inventory Turnover" outputs at the bottom.



INTERACTIVE ILLUSTRATION 3 Inventory Turnover and Days in Inventory



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp219pGVd



Applying the data from Exhibit 1 gives us the following:

$$\begin{aligned} &\text{Prada's average inventory in 2015} \\ &= \frac{(\text{beginning inventory} + \text{ending inventory})}{2} \\ &= \frac{(\text{€}450 + \text{€}655)}{2} \\ &= \text{€}552 \end{aligned}$$

$$\text{Prada's inventory turnover in 2015} = \frac{\text{€}1,001}{\text{€}552} = 1.8$$

Similarly, we calculate inventory turnover for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Inventory Turnover	5.3	4.3	5.1	1.8	6.4

As a fast fashion business, it is no surprise that Urban Outfitters turns over its inventory faster (6.4 times a year) than Prada's luxury products (1.8 times). Urban

Outfitters is also somewhat more efficient in handling its inventory when compared to the other fast fashion business, Gap, whose inventory turnover is 5.3.

Another metric for measuring inventory use is the **days inventory held**, the average number of days the inventory is held before it is sold, as opposed to the inventory turnover, which measures how many times the inventory turned over during the period. It is calculated dividing the COGS by 365 days, then dividing the average inventory by this number.

$$\text{Days inventory held} = \frac{\text{average inventory}}{\text{cost of goods sold}/365 \text{ days}}$$

or:

$$\text{Days inventory held} = \frac{365}{\text{inventory turnover}}$$

Applying the data for our Prada example yields the following:

$$\text{Prada's days inventory held in 2015} = \frac{365}{\text{€1,001}/\text{€552}} = 201$$

Similarly, we calculate days inventory held for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Days Inventory Held	69	86	72	201	57

Again it makes sense that Urban Outfitters turns its inventory faster than Prada because the average number of days that Urban Outfitters holds its inventory is substantially shorter—57 days versus Prada’s 201 days.

2.3.4 Accounts Receivable Turnover

Accounts receivable (AR) turnover measures a company’s efficiency in collecting receivables from its customers. Companies often provide credit to their customers as a way to increase sales. However, this practice comes at the risk of having some customers be unable to pay when the bill comes due. The accounts receivable turnover is calculated by dividing sales revenues by the average accounts receivable (AR) balance. While AR turnover depends on the company’s credit practices, a

declining AR turnover indicates a weaker ability to collect cash from customers, which could indicate a business weakness. For example, it could mean that the company's customers are having trouble paying what they owe.

$$\text{Accounts receivable turnover} = \frac{\text{sales}}{\text{average accounts receivable}}$$

$$\begin{aligned} &\text{Prada's average accounts receivable (AR) in 2015} \\ &= \frac{(\text{beginning AR} + \text{ending AR})}{2} \\ &= \frac{(\text{€308} + \text{€346})}{2} \\ &= \text{€327} \end{aligned}$$

$$\text{Prada's accounts receivable turnover in 2015} = \frac{\text{€3,552}}{\text{€327}} = 10.9$$

Similarly, we calculate the AR turnover for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Accounts Receivable Turnover	44.6	54.3	6.0	10.9	52.9

A related metric is **days sales outstanding (DSO)**, also referred to as the **average collection period** or days sales in receivables. It measures the average number of days it takes for a business to collect payment from a customer. Say the firm's credit policy is to allow payment within 30 days. If, for example, the DSO is 40 days, it may indicate that there is a payment problem. The DSO is calculated by dividing the average accounts receivable by the sales per day, that is, sales divided by 365.

$$\text{Days sales outstanding} = \frac{\text{average accounts receivable}}{\text{sales}/365 \text{ days}}$$

or

$$\text{Days sales outstanding} = \frac{365 \text{ days}}{\text{accounts receivable turnover}}$$

Using the data from Exhibit 1, we find the following:

$$\text{Prada's days sales outstanding in 2015} = \frac{365}{\text{€3,552}/\text{€327}} = 34$$

Similarly, we calculate DSO for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Days Sales Outstanding	8	7	61	34	7

Retailers such as the ones in our examples typically do not offer credit to their customers, therefore they do not incur high levels of receivables. Gap, Inditex, and Urban Outfitters' low level of DSO generally results from the companies' vendors of miscellaneous goods (e.g., furniture and fixtures) and sundry services. Therefore, Nordstrom's DSO of about two months and Prada's of about one month deserve some attention. It turns out that Nordstrom has a credit card operation that offers Nordstrom-branded cards to its customers, which retains the receivables on its balance sheet. This accounts for Nordstrom's lower receivable turnover ratio and higher DSO. Prada, on the other hand, sells not only through its own stores but also through franchises. Therefore, the franchises owe Prada for the products it sells to them for sale to the end customer, as well as possible royalty fees.

2.3.5 Accounts Payable Turnover

The **accounts payable (AP) turnover** measures how long it takes a company to pay its vendors, including suppliers of inventory, services, or other noninventory items. Recall that payables are the flip side of receivables, that is, payables occur when the vendor allows the company credit in paying for goods or services rendered. Accounts payable turnover is calculated as total purchases divided by the average accounts payable. (An alternate definition substitutes cost of goods sold for purchases).

$$\text{Accounts payable turnover} = \frac{\text{purchases}}{\text{average accounts payable}}$$

Applying this ratio to our Prada example gives us the following:

$$\begin{aligned}\text{Prada's purchases in 2015} &= \text{COGS} + \text{ending inventory} - \text{beginning inventory} \\ &= \text{€1,001} + \text{€655} - \text{€450} = \text{€1,206}\end{aligned}$$

$$\begin{aligned}
 &\text{Prada's average accounts payable (AP) in 2015} \\
 &= \frac{(\text{beginning AP} + \text{ending AP})}{2} \\
 &= \frac{(\text{€349} + \text{€437})}{2} \\
 &= \text{€393}
 \end{aligned}$$

$$\text{Prada's accounts payable turnover in 2015} = \frac{\text{€1,206}}{\text{€393}} = 3.1$$

Similarly, we calculate accounts payable turnover for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Accounts Payable Turnover	8.4	3.2	6.6	3.1	15.0

As with inventory and receivables, it is useful to think of accounts payable turnover in terms of the number of days it takes the company to pay its accounts payable, or **days payable outstanding (DPO)**. This ratio is calculated by dividing the average accounts payable by the average daily purchases (that is, total purchases divided by 365).

$$\text{Days payable outstanding} = \frac{\text{average accounts payable}}{\text{purchases}/365 \text{ days}}$$

Alternatively,

$$\text{Days payable outstanding} = \frac{365 \text{ days}}{\text{accounts payable turnover}}$$

Applying the data from Exhibit 1 gives us the following:

$$\text{Prada's days payable outstanding in 2015} = \frac{365}{\text{€1,206}/\text{€393}} = 119$$

Similarly, we calculate DPO for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Days Payable Outstanding	44	115	56	119	24

The difference between Urban Outfitters' days payable outstanding of 24 days versus Prada's 119 days indicates that Urban Outfitters pays its vendors much faster than Prada does. To evaluate the efficiency implication of these different days payable outstanding ratios, we would need to compare the historical numbers as well as the credit terms. Credit from suppliers serves as a form of short-term financing that can be beneficial if it is cheaper than bank credit or other sources of financing. However, an increasing trend in days payable outstanding can suggest that the company has liquidity problems, making it unable to pay its suppliers on time.

2.3.6 Cash Conversion Cycle

The length of time between when a company must pay its suppliers for inventory until it collects cash from its customers is called the **cash conversion cycle**. It is the sum of days inventory held plus days sales outstanding less days payable outstanding. A negative cash conversion cycle typically indicates that the business purchased goods from a supplier on credit and can sell the inventory and collect cash from the customer even before the supplier requires payment to be made.

Cash conversion cycle = days inventory + days sales outstanding – days payable outstanding

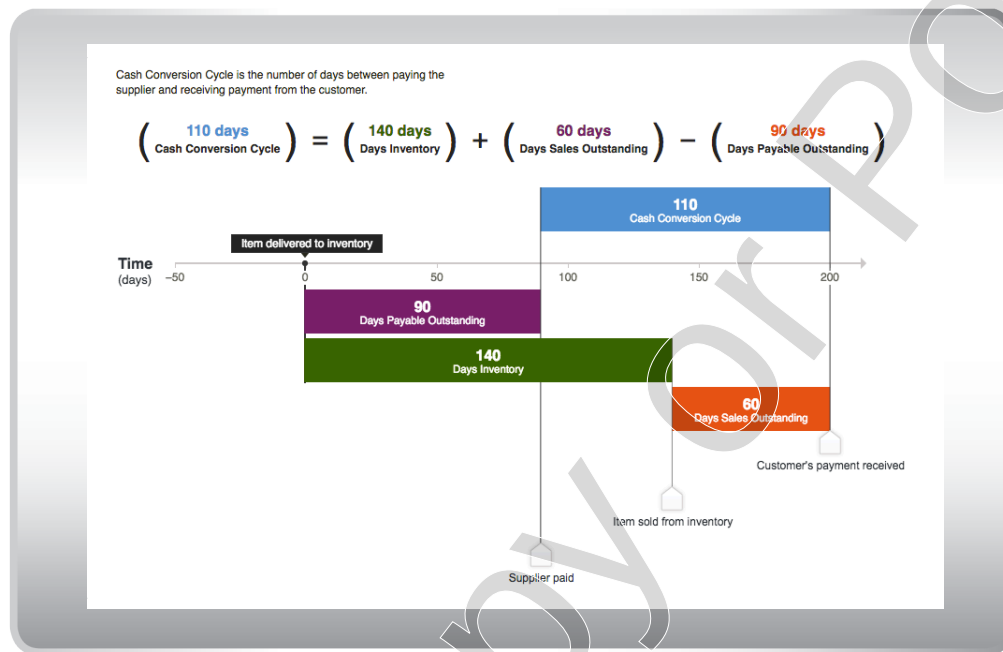
The pertinent events that affect the cash conversion cycle—payment (for an item) to a supplier, the item (or a product that includes that item) is sold, and the customer's payment for that sale is received—are displayed on a timeline in **Interactive Illustration 4**. Use the sliders to change the timing of these events, to push the cash conversion cycle to be long, short, or negative.



INTERACTIVE ILLUSTRATION 4 Cash Conversion Cycle



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2ujeHXe



Using the data for Prada in Exhibit 1, we find the following:

$$\text{Prada's cash conversion cycle in 2015} = 201 + 34 - 119 = 116$$

Similarly, we calculate the cash conversion cycle for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Cash Conversion Cycle	33	(22)	77	116	39

Urban Outfitters' cash conversion cycle of 39 days means the company must finance 39 days' worth of sales, compared to Prada's 116 days. Inditex, on the other hand, has a negative cash conversion cycle of 22 days; the company can sell its goods before paying its suppliers, which means it does not need to invest in financing its inventory. This is because Inditex takes longer to pay its suppliers (days payable outstanding = 115 days), not because of its receivables days outstanding (days sales outstanding = 7) or the number of days it holds inventory (days inventory held = 86). Prada, in comparison, holds its inventory longer and, as a result, has more cash tied up in its business operations.

2.3.7 Working Capital Turnover

Closely related to cash conversion is the concept of working capital, defined as current assets less current liabilities. **Working capital turnover** is calculated by dividing net sales by the result of average current assets minus average current liabilities.

$$\text{Working capital turnover} = \frac{\text{sales}}{\text{average working capital}}$$

where

$$\text{Working capital} = \text{current assets} - \text{current liabilities}$$

For Prada, this result is as follows:

$$\begin{aligned} & \text{Prada's average current assets in 2015} \\ &= \frac{(\text{beginning current assets} + \text{ending current assets})}{2} \\ &= \frac{(\text{€1,461} + \text{€1,900})}{2} \\ &= \text{€1,681} \end{aligned}$$

$$\begin{aligned} & \text{Prada's average current liabilities in 2015} \\ &= \frac{(\text{beginning current liabilities} + \text{ending current liabilities})}{2} \\ &= \frac{(\text{€707} + \text{€1,115})}{2} \\ &= \text{€911} \end{aligned}$$

$$\text{Prada's working capital turnover in 2015} = \frac{\text{€3,552}}{(\text{€1,681} - \text{€911})} = 4.6$$

Similarly, we calculate the working capital turnover for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Working Capital Turnover	8.1	5.4	5.3	4.6	5.9

Working capital turnover measures the efficiency with which a company manages its working capital to generate sales. Gap with a working capital turnover of 8.1 is the most effective of the retailers listed.

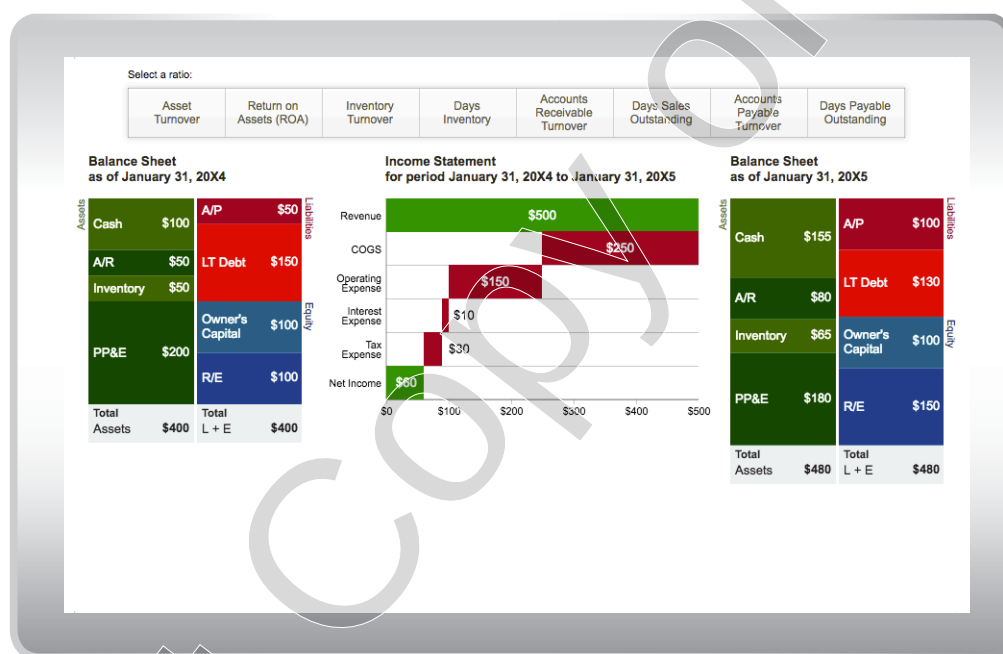
Calculations for common efficiency measures are visualized in **Interactive Illustration 5**. Select a metric from the top row to see what items are drawn from the income statement and balance sheet to calculate that ratio.



INTERACTIVE ILLUSTRATION 5 Efficiency Ratios



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2uo8MQH



2.4 Financial Leverage

Having examined the drivers of profitability and operating efficiency, we next assess how capital structure affects a company's performance. This leads to the last component of the DuPont decomposition framework, financial leverage. In the DuPont decomposition formula, the last term is also referred to as the **equity multiplier** and is calculated by dividing average total assets by average shareholder's equity.

Managers can successfully use financial leverage to improve a company's performance if the return on borrowed funds is greater than the interest cost of those borrowed funds on an after-tax basis because interest cost is tax-deductible. The benefit of borrowing is that managers can operate from a larger asset base than the amount that would be financed only from shareholders' equity. For example, if a firm

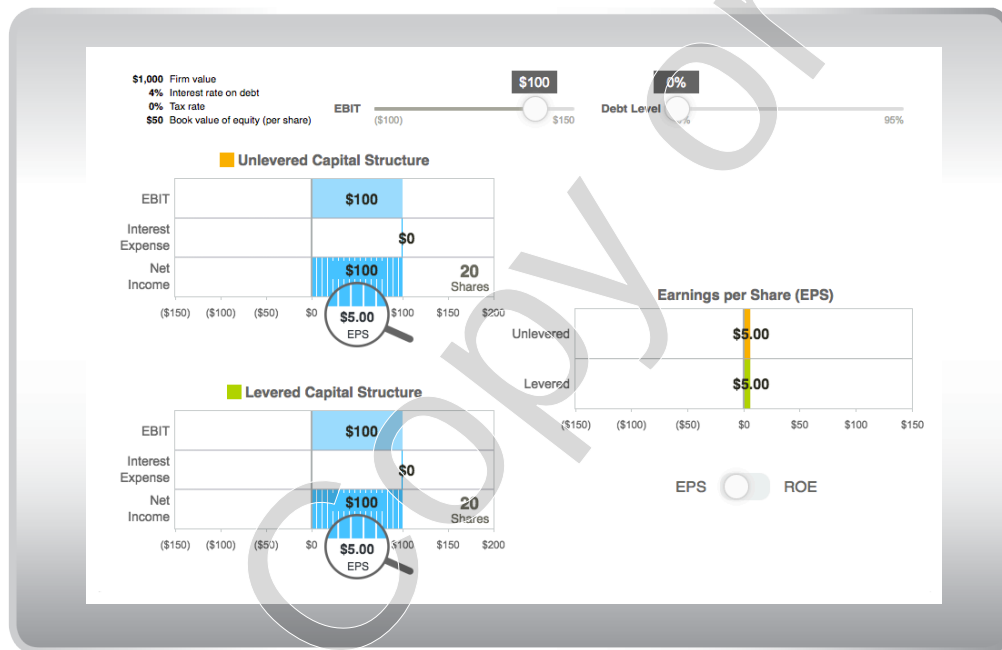
receives a \$100 equity investment and has no debt, it can use that \$100 to buy assets that generate revenues. If the firm borrows an additional \$100 as debt, it can use the extra capital to purchase more revenue-generating assets. This would result in greater profits for shareholders as long as the business generates a higher return on its assets than what it must pay to service the debt. **Interactive Illustration 6** shows how leverage affects companies' EPS and ROE.



INTERACTIVE ILLUSTRATION 6 Leverage Ratios



Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2GiuO8Y



The flip side of leverage is that the company becomes committed to meeting interest payments, whereas dividend payment to equity holders is discretionary. Borrowing can thus create a risk of financial distress when the firm's performance is depressed. If the business were to suffer a loss, the debt amplifies the loss's effect on equity holders. Therefore, companies should avoid very high levels of debt financing. Analysts use the extent of financial leverage to assess the leverage's benefit as well as to measure distress risk.

Using the ROE decomposition formula provided earlier, we can measure the impact of all non-equity financing—or any kind of debt—on the firm's ROE. If all the assets are financed by equity, the multiplier is 1. As debt increases, the equity multiplier also increases.

$$\text{Equity multiplier} = \frac{\text{average total assets}}{\text{average equity}}$$

Using the data in Exhibit 1, we calculate the following:

$$\begin{aligned}
 &\text{Prada's average total assets in 2015} \\
 &= \frac{(\text{beginning total assets} + \text{ending total assets})}{2} \\
 &= \frac{(\text{€3,888} + \text{€4,739})}{2} \\
 &= \text{€4,314}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Prada's average shareholders' equity (SE) in 2015} \\
 &= \frac{(\text{beginning SE} + \text{ending SE})}{2} \\
 &= \frac{(\text{€2,688} + \text{€3,001})}{2} \\
 &= \text{€2,845}
 \end{aligned}$$

$$\text{Prada's equity multiplier in 2015} = \frac{\text{€4,314}}{\text{€2,845}} = 1.5$$

Similarly, we calculate the equity multiplier for the other retailers for fiscal year 2015 as follows:

<i>FY 2015</i>	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Equity Multiplier	2.6	1.5	3.9	1.5	1.4

Urban Outfitters and Prada's equity multipliers are quite similar—1.4 and 1.5, respectively—indicating that the two companies have similar capital structure strategies. However, Nordstrom has a high equity multiplier of 3.9. Recall that Nordstrom's ROE was 31.9%, significantly higher than Prada's 15.8%, despite Nordstrom's lower ROA (8.1%) compared to Prada's (10.4%). The answer lies in the significantly higher level of leverage as indicated by Nordstrom's equity multiplier, which is more than double Prada's. Can Prada improve its returns to shareholder equity (ROE) by taking on more debt? Prada's managers must grapple with this question. The answer would depend on the extent of financial risk they wish to take on. Alternately, should Nordstrom's shareholders be concerned that the company has very high debt levels? They would need to consider whether Nordstrom's business operations can support these high debt levels without causing the firm to slip into financial distress.

Another commonly used measure of leverage is the debt-to-equity ratio, which is calculated as the ratio of average total liability divided by average shareholder's equity.

$$\text{Leverage ratio} = \frac{\text{average total liabilities}}{\text{average shareholders' equity}}$$

Using the data in Exhibit 1, we find the following:

$$\begin{aligned} &\text{Prada's average total liabilities in 2015} \\ &= \frac{(\text{beginning total liabilities} + \text{ending total liabilities})}{2} \\ &= \frac{(\text{€1,187} + \text{€1,721})}{2} \\ &= \text{€1,454} \end{aligned}$$

$$\begin{aligned} &\text{Prada's average shareholders' equity (SE) in 2015} \\ &= \frac{(\text{beginning SE} + \text{ending SE})}{2} \\ &= \frac{(\text{€2,688} + \text{€3,001})}{2} \\ &= \text{€2,845} \end{aligned}$$

$$\text{Prada's leverage ratio in 2015} = \frac{\text{€1,454}}{\text{€2,845}} = 0.5$$

Similarly, we calculate the leverage ratio for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Leverage Ratio	1.6	0.5	2.9	0.5	0.4

Not surprisingly, Nordstrom's leverage ratio, at 2.9, is significantly higher than Prada's at 0.5. We can peel the onion further and compare the different retailers in terms of long-term debt versus short-term liabilities.

$$\text{Long-term leverage ratio} = \frac{\text{average long-term liabilities}}{\text{average equity}}$$

$$\begin{aligned}
 &\text{Prada's average long-term (LT) liabilities in 2015} \\
 &= \frac{(\text{beginning LT liabilities} + \text{ending LT liabilities})}{2} \\
 &= \frac{(\text{€480} + \text{€606})}{2} \\
 &= \text{€543}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Prada's average shareholders' equity (SE) in 2015} \\
 &= \frac{(\text{beginning SE} + \text{ending SE})}{2} \\
 &= \frac{(\text{€2,688} + \text{€3,001})}{2} \\
 &= \text{€2,845}
 \end{aligned}$$

$$\text{Prada's long-term leverage ratio in 2015} = \frac{\text{€543}}{\text{€2,845}} = 0.2$$

Similarly, we calculate the LT leverage ratio for the other retailers for fiscal year 2015 as follows:

<i>FY 2015</i>	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Long-Term Leverage Ratio	0.8	0.1	1.8	0.2	0.1

These ratios indicate that Nordstrom has significantly greater long-term debt compared to its equity than the other retailers in our sample do.

2.4.1 Interest Coverage Ratio

As discussed earlier, one possible result of higher leverage is the risk that the company may not be able to meet its debt payment obligations. Many debt providers, such as banks, protect themselves from their creditors' default by requiring the companies to follow certain restrictions (or covenants) defined by the ratios that measure the extent of financing risk. The **interest coverage ratio** is used to gauge whether a business can make the interest payments on its outstanding debt. The ratio is measured as earnings before interest and taxes (EBIT) divided by interest expense.

$$\text{Interest coverage ratio} = \frac{\text{EBIT}}{\text{interest expense}}$$

which for Prada would be as follows:

$$\text{Prada's interest coverage ratio in 2015} = \frac{\text{€702}}{\text{€13}} = 54$$

Similarly, we calculate the interest coverage ratio for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Interest Coverage Ratio	28	N/A	11	54	N/A

Interest coverage ratio is useful for assessing the firm's long-term solvency by examining its ability to cover interest expense on debt, typically long-term debt. Nordstrom, with an interest coverage ratio of 11, has the lowest ability to meet its debt obligations within this sample of retailers. Although it is difficult to assess the extent of distress risk from this ratio alone, it is clear that Prada, with a ratio of 54, and Gap at 28, have a greater amount of earnings to cover interest expense. Because Urban Outfitters and Inditex did not incur much interest expense in 2015, their interest coverage ratios are less meaningful.

2.4.2 Liquidity Ratios

Analysts, especially those concerned with assessing a company's short-term liquidity (for example, a credit analyst in a bank), use measures such as the **current ratio** to determine if funds provided by current assets would be sufficient to meet demand from current liabilities.

The current ratio is calculated as current assets divided by current liabilities. Current assets include cash as well as assets such as inventory and accounts receivable that are likely to convert to cash in the short term. Current liabilities include accounts payable, wages payable, and other items that will require cash payout in the short term. The higher the current ratio, the easier it is for the business to meet its approaching liabilities. However, if the current ratio is too high, it might indicate that the business is not managing its working capital efficiently.

$$\text{Current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

For Prada, the current ratio is as follows:

$$\text{Prada's current ratio in 2015} = \frac{\text{€1,900}}{\text{€1,115}} = 1.7$$

Similarly, we calculate current ratio for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Current Ratio	1.9	1.9	1.9	1.7	2.3

Overall, all the retailers' current ratios hover around similar levels. Banks and other short-term lenders often impose requirements for the current ratio level and that of other, similar liquidity ratios in a given industry. It is also helpful to compare current with historical levels of the ratio to determine whether there are any reasons for concern or whether this metric should be differentiated across similar companies.

The **quick ratio**, or acid test ratio, is similar to the current ratio except only highly liquid current assets are considered available for paying current liabilities. This ratio measures a firm's ability to meet current obligations even if its inventory cannot be sold immediately. It is calculated by considering cash, marketable securities, and accounts receivable, but not inventory, which is relatively less liquid; businesses in distress may have trouble turning their inventory into cash.

$$\text{Quick ratio} = \frac{(\text{cash and marketable securities} + \text{accounts receivable})}{\text{current liabilities}}$$

Applying the data from Exhibit 1, we get the following:

$$\text{Prada's quick ratio in 2015} = \frac{\text{€715} + \text{€346}}{\text{€1,115}} = 1.0$$

Similarly, we calculate the quick ratio for the other retailers for fiscal year 2015 as follows:

FY 2015	Gap	Inditex	Nordstrom	Prada	Urban Outfitters
Quick Ratio	0.8	1.3	1.1	1.0	0.9

The retailers in our sample generally appear to be quite comfortable in meeting their current liabilities from highly liquid current assets though Gap and Urban Outfitters are a little less so than the other companies.

3 SUMMARY

The financial analysis framework provides a methodology for assessing and digging deeper into the drivers of a company's performance. Analysts and others can use financial ratios to compare the company's performance to its peers or to its own historical performance. Managers can then use the insights to improve the way the business operates, by improving either its profit margins, its asset utilization, or its capital structure. In this reading, we have used the DuPont decomposition framework to provide a basic structure for ratio analysis. There are many additional ratios not covered here, and you may find somewhat different definitions from what we have used here.

In order to conduct insightful financial analysis, you must develop a deep understanding of the business economics of a company and its industry to select the appropriate ratios to measure and then to interpret them correctly. As seen in our retail examples, it is also important to probe deeper to understand the causes of differences in the ratios' measures. A good analyst typically uses a portfolio of ratios to understand the overall picture of a company and the drivers of its performance.

EXHIBIT 1 Prada Group's Balance Sheet and Income Statement

Prada Group: Consolidated statement of financial position		
(amounts in millions of Euros)	January 31, 2015	January 31, 2014
Assets		
Current assets		
Cash and marketable securities	715	582
Trade receivables, net	346	308
Inventory, net	655	450
Other current assets	184	121
Total current assets	1,900	1,461
Non-current assets		
Property, plant, and equipment	1,474	1,230
Intangible assets	943	901
Other non-current assets	422	296
Total non-current assets	2,839	2,427
Total assets	4,739	3,888
Liabilities and shareholders' equity		
Current liabilities		
Bank overdrafts and short-term loans	267	67
Trade payable	437	349
Other current liabilities	411	291
Total current liabilities	1,115	707
Non-current liabilities		
Long-term financial payables	255	208
Deferred tax liabilities	42	42
Other non-current liabilities	309	230
Total non-current liabilities	606	480
Total liabilities	1,721	1,187
Share capital		
Share capital	256	256
Other reserves	2,163	1,853
Translation reserves	131	(49)
Net income for the year	451	628
Total shareholders' equity	3,001	2,688
Non-controlling interests	17	13
Total liabilities and shareholders' equity	4,739	3,888

Consolidated income statement		<i>For the twelve months ended</i>	
(amounts in millions of Euros)		January 31, 2015	January 31, 2014
Net revenue		3,552	3,587
Cost of goods sold		(1,001)	(939)
Gross margin		2,551	2,648
Selling, general and administrative costs		1,716	1,580
Product design and development costs		133	129
Operating expenses		1,849	1,709
Earnings before interest and taxes		702	939
Net interest expenses		(13)	(9)
Other income/(expenses)		(21)	(8)
Income before taxes		668	922
Taxation		(209)	(285)
Net income for the year		459	637
Net income—non-controlling interests		8	10
Net income—shareholders		451	627
Basic and diluted earnings per share in euros per share		0.176	0.245

Source: Extracted from Prada Annual Report for the year ended January 31, 2015.

4 SUPPLEMENTAL READING

4.1 Key Ratios

$$\text{Accounts payable turnover} = \frac{\text{purchases}}{\text{average accounts payable}}$$

$$\text{Accounts receivable turnover} = \frac{\text{sales}}{\text{average accounts receivable}}$$

$$\text{Asset turnover} = \frac{\text{sales}}{\text{average total assets}}$$

$$\text{Cash conversion cycle} = \text{days inventory} + \text{days sales outstanding} - \text{days payable outstanding}$$

$$\text{Cost of goods sold} = \text{beginning inventory} + \text{purchases} - \text{ending inventory}$$

$$\text{Credit purchases} = \text{cost of goods sold} + \text{ending inventory} - \text{beginning inventory}$$

$$\text{Current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

$$\text{Days inventory held} = \frac{\text{average inventory}}{\text{cost of goods sold}/365 \text{ days}}$$

$$\text{Days inventory held} = \frac{365}{\text{inventory turnover}}$$

$$\text{Days payable outstanding} = \frac{\text{average accounts payable}}{\text{purchases}/365 \text{ days}}$$

$$\text{Days payable outstanding} = \frac{365 \text{ days}}{\text{accounts payable turnover}}$$

$$\text{Days sales outstanding} = \frac{\text{average accounts receivable}}{\text{sales}/365 \text{ days}}$$

$$\text{Days sales outstanding} = \frac{365 \text{ days}}{\text{accounts receivable turnover}}$$

$$\text{Equity multiplier} = \frac{\text{average total assets}}{\text{average equity}}$$

$$\text{Gross profit margin} = \frac{\text{gross profit}}{\text{sales}}$$

$$\text{Interest coverage ratio} = \frac{\text{EBIT}}{\text{interest expense}}$$

$$\text{Inventory turnover} = \frac{\text{cost of goods sold (COGS)}}{\text{average inventory}}$$

$$\text{Leverage ratio} = \frac{\text{average total liabilities}}{\text{average shareholders' equity}}$$

$$\text{Long-term asset turnover} = \frac{\text{sales}}{\text{average long-term assets}}$$

$$\text{Long-term leverage ratio} = \frac{\text{average long-term liabilities}}{\text{average equity}}$$

$$\text{Net profit margin} = \frac{\text{net income}}{\text{sales}}$$

$$\text{Quick ratio} = \frac{(\text{cash and marketable securities} + \text{accounts receivable})}{\text{current liabilities}}$$

$$\text{ROA} = \frac{\text{net income}}{\text{average total assets}}$$

$$\text{ROE} = \frac{\text{net income}}{\text{sales}} \times \frac{\text{sales}}{\text{total assets}} \times \frac{\text{total assets}}{\text{equity}} = \frac{\text{net income}}{\text{equity}}$$

$$\text{SG\&A margin} = \frac{\text{selling, general, and administrative (SG\&A) expenses}}{\text{sales}}$$

$$\text{Working capital} = \text{current assets} - \text{current liabilities}$$

$$\text{Working capital turnover} = \frac{\text{sales}}{\text{average working capital}}$$

5 KEY TERMS

accounts payable (AP) turnover Measures how long it takes a company to pay its vendors, including suppliers of inventory, services, or other noninventory items.

accounts receivable (AR) turnover Measures a company's efficiency in collecting receivables from its customers.

asset turnover Measures how efficiently the company uses its assets to generate return.

average collection period Measures the average number of days it takes for a business to collect payment from a customer. Also called days sales outstanding (DSO).

balance sheet A financial statement that provides a snapshot of the company's resources and the claims on those resources at a specific point in time.

cash conversion cycle The length of time between when a company must pay its suppliers for inventory until it collects cash from its customers.

current ratio Measures the sufficiency of a firm's funds, provided by current assets, to meet demand from current liabilities.

days inventory held The average number of days the inventory is held before it is sold.

days payable outstanding (DPO) The number of days it takes the company to pay its accounts payable.

DuPont framework A ratio analysis method, originated by the DuPont Corporation in the 1920s, which decomposes ROE into subcomponents to provide a deeper look at how profit (or loss) was generated.

equity multiplier The measure of the impact of debt on a firm's ROE.

financial leverage The degree to which a company uses fixed-cost financing such as bank loans, bonds, and preferred stock. The more debt and preferred stock financing a company uses, the higher its financial leverage.

gross profit margin The amount of profit that is left to cover other expenses after only the cost of goods sold is subtracted from revenues.

income statement A financial statement that shows an entity's operating performance over a given period of time.

interest coverage ratio Gauges whether a business can make the interest payments on its outstanding debt.

inventory turnover Measures how often the inventory is sold during a given time period.

profitability In the DuPont framework, the profit margin that the company achieves from each dollar of sales after all expenses have been accounted for.

quick ratio Measures a firm's ability to meet current obligations even if its inventory cannot be sold immediately.

ratio analysis The use of simple calculations as financial analysis tools, which can be used to gain insights about a company's business model and financial performance.

return on assets (ROA) An indicator of the company's profitability relative to its assets or total capital employed in the firm.

return on equity (ROE) A measure of a business's profitability in relation to its shareholder equity. It is found by dividing net income by the book value of the equity.

selling, general, and administrative (SG&A) margin The amount of SG&A expenses incurred by a company for every dollar of revenues earned.

shareholders' equity A measure of a business's profitability that is found by subtracting liabilities from the company's assets. It consists of contributed capital (stock) and retained earnings.

statement of cash flows A financial statement showing cash sources and uses by a company over an accounting cycle.

working capital The liquid capital used in day-to-day business operations.

working capital turnover Measures the efficiency with which a company manages its working capital to generate sales.

6 ENDNOTES

- 1 Inditex, Annual Report 2014 (A Coruña, Spain: Inditex, 2015) p. 193.
- 2 Prada Group, Annual Report 2014 (Milan, Italy, 2015), p. 5.
- 3 Prada Group, Annual Report 2014 (Milan, Italy, 2015), p. 17.
- 4 Urban Outfitters, Inc. SEC Form 10-K, January 31, 2015 (Philadelphia, PA), p. 1.
- 5 Urban Outfitters, Inc. SEC Form 10-K, January 31, 2015 (Philadelphia, PA), pp. 2–3.
- 6 Nordstrom, Inc., SEC Form 10-K, January 31, 2015 (Seattle, WA), p. 4.

7 INDEX

- accounts payable (AP) turnover, 18, 19, 32, 34
- accounts receivable (AR) turnover, 16, 17, 32, 34
- asset turnover, 11, 12, 32, 34
- assets, in return on equity calculation, 6, 7
- average collection period, 17, 34
- balance sheet, 4, 34
- cash conversion cycle, 20, 21, 32, 35
- cost of goods sold, 32
- credit purchase, 32
- current ratio, 28, 29, 32, 35
- days inventory held, 16, 32, 35
- days payable outstanding (DPO), 19, 20, 21, 32, 33, 35
- days sales outstanding (DSO), 17, 18, 20, 21, 32, 33, 35
- DuPont framework, 3, 5, 7, 35
- efficiency ratios, 23
- equity multiplier, 23, 24, 25, 33, 35
- equity, in return on equity calculation, 6, 7
- financial analysis, 3, 5, 7
- financial leverage, 6, 35
- Gap, 3, 4, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 25, 26, 27, 28, 29, 30
- gross profit margin, 9, 33, 35
- income statement, 4, 35
- Inditex, 3, 4, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 21, 22, 25, 26, 27, 28, 29
- interest coverage ratio, 27, 28, 33, 35
- interest expense, 13
- inventory turnover, 14, 15, 33, 35
- leverage, 35
- leverage ratio, 26, 33
- liquidity ratios, 28
- long-term asset turnover, 13, 33
- long-term leverage ratio, 26, 27, 33
- net income, 6, 7, 12, 13
- net profit margin, 8, 33
- Nordstrom, 3, 4, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 21, 22, 25, 26, 27, 28, 29
- operating efficiency, 6, 11
- performance, analysis of, 3, 5, 7, 9, 10
- performance, financial leverage and, 23, 24, 30
- performance, mix of assets driving, 13
- Prada, 3, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 25, 26, 27, 28, 29, 31
- profit margins, 6
- profitability, 6, 8, 35
- profitability ratios, 10
- quick ratio, 29, 34, 35
- ratio analysis, 3, 35
- ratios, periods used in calculating, 5
- return on assets (ROA), 5, 12, 13, 34, 36
- return on equity (ROE), 5, 6, 7, 8, 34, 36
- sales, in return on equity calculation, 6, 7
- selling, general, and administrative (SG&A) margin, 9, 10, 34, 36
- shareholders' equity (SE), 5, 7, 23, 25, 26, 27, 31, 33, 36
- statement of cash flows, 4, 36
- Urban Outfitters, 3, 4, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 25, 26, 27, 28, 29, 30
- working capital, 14, 22, 34, 36
- working capital turnover, 22, 34, 36